Reversal Chronology From a 7.9- to 11.5-M.Y.-Old Volcanic Sequence in Central Arizona: Comparison With Ocean Floor Polarity Record

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A magnetic reversal chronology has been obtained from a section over 700 m thick of radiometrically dated lava flows (7.9–11.5 m.y.) from the Hackberry Mountain area in central Arizona. There are an adequate number of polarity reversals in the Hackberry Mountain volcanic sequence to allow an apparent correlation with the sea floor record assigned to this time interval. A correlation using a few radiometric ages widely spaced in the section for temporal control proves to be incorrect because it does not assess the length of hiatuses in the terrestrial sequence and assumes that the rate of volcanism and sedimentation is uniform. When a large number of stratigraphically controlled radiometric ages are used, temporal compression, extension, and shifting of parts of the Hackberry Mountain reversal pattern produce a different correlation with the ocean floor reversal pattern. In spite of such adjustments, not all of the stratigraphically controlled K-Ar ages and associated polarity reversals are in accord with ages estimated for polarity boundaries in the sea floor record. The Hackberry Mountain pattern can be less well fitted to the marine sediment polarity record than to the sea floor anomaly record, implying that there are unrecognized hiatuses in the marine sediment record. We conclude that a lack of apparent similarity in the adjusted 8- to 11-m.y.-old Hackberry Mountain pattern and the currently accepted reversal time scale derived from marine magnetic anomalies reflects (1) loss or absence of the geologic record in the terrestrial volcanic sequence, (2) perhaps brief reversals not recognized in the marine record, and (3) a possible lack of strict linearity in the assignment of estimated ages to the sea floor pattern, which is based on the assumption of uniform spreading rates. A reversal pattern obtained from a terrestrial volcanic sequence older than about 5 m.y. thus is of no value, in itself, for establishing a chronology.

INTRODUCTION

This study was started in 1970 with two objectives in mind: (1) to see if the terrestrial magnetic reversal time scale could be extended beyond the 4.5-m.y. limit established by *Cox et al.* [1963, 1968] and *Cox* [1969] by employing a stratigraphically controlled, radiometrically dated sequence of terrestrial lava flows, and (2) to ascertain if a magnetic reversal pattern in a sequence of terrestrial lavas 5 m.y. or older could be used to date the rocks. The results produced a valuable lesson about the limitations of magnetic reversal patterns and K-Ar chronology used on terrestrial volcanic sequences.

The rocks selected for the study include part of the sequence of basalt and andesite lava flows, dacite tuffs, and interbedded sedimentary strata that border the south end of the Verde Valley, northwest of Hackberry Mountain in central Arizona (Figure 1). The composite sequence studied is about 700 m thick. Several preliminary K-Ar age determinations on rocks near the top and bottom of the section studied indicated that the sequence was deposited in the time interval of about 7–12 m.y. ago. Radiometric dates from elsewhere in the region suggested that a longer time span might be available if correlations could be established between the Verde Formation (about 7.5–3 m.y. old; *Bressler and Butler* [1978]) and the sequence of rocks near Hackberry Mountain. Moreover, the time interval might also be extended back to about 14 m.y. if correlations could be made with basalt 11–14 m.y. old known to exist elsewhere in the region.

GENERAL GEOLOGY

Detailed geologic mapping along and near the Verde River at the south end of the Verde Valley (Verde River section, Figure 1) in the Hackberry Mountain area has resulted in the composite stratigraphic section shown in Figure 2. The section comprises, in ascending order, the Hickey Formation, the Thirteenmile Rock Volcanics of *Elston et al.* [1974], and the Verde Formation. Olivine basalt flows dominate the section. They range from single simple flows a few meters thick to compound flows a few tens of meters thick. The flows issued from fissures and basaltic cones scattered throughout the region, and they mask Precambrian and Paleozoic bedrock from Flagstaff on the Colorado Plateau to the north, to the Cave Creek area of the desert region of Phoenix to the south, a distance of 240 km.

The Hickey Formation, named for exposures at Hickey Mountain [*Anderson and Creasey,* 1958] in the Black Hills that border the west side of the Verde Valley, consists of Tertiary sedimentary rocks overlain by basaltic flows. These flows cap much of the Black Hills, are found on the rim of the Colorado Plateau to the north, and can be traced south to the area of Cave Creek. Basalt of the Hickey Formation was emplaced in the interval of about 11–14.5 m.y. ago [*Damon,* 1965, 1968; *McKee and Anderson,* 1971; *McKee and McKee,* 1972]. The distribution of dated flows in central Arizona is summarized by *Elston et al.* [1974], who reported preliminary radiometric ages for the Hackberry Mountain section and for a stratigraphically higher section of basalt and rhyolitic deposits in the area of Thirteenmile Rock (Figure 1). In the Verde River section northwest of Hackberry Mountain, basalt of the Hickey Formation in the lower part of the section (Figure 2) was tentatively correlated on the basis of mapping with basalt flows of the Hickey that cap the adjacent Black Hills.

The Thirteenmile Rock Volcanics of *Elston et al.* [1974] overlies the Hickey Formation. The Thirteenmile Rock Volcanics consist of dacite tuff and flows derived from the Hack-

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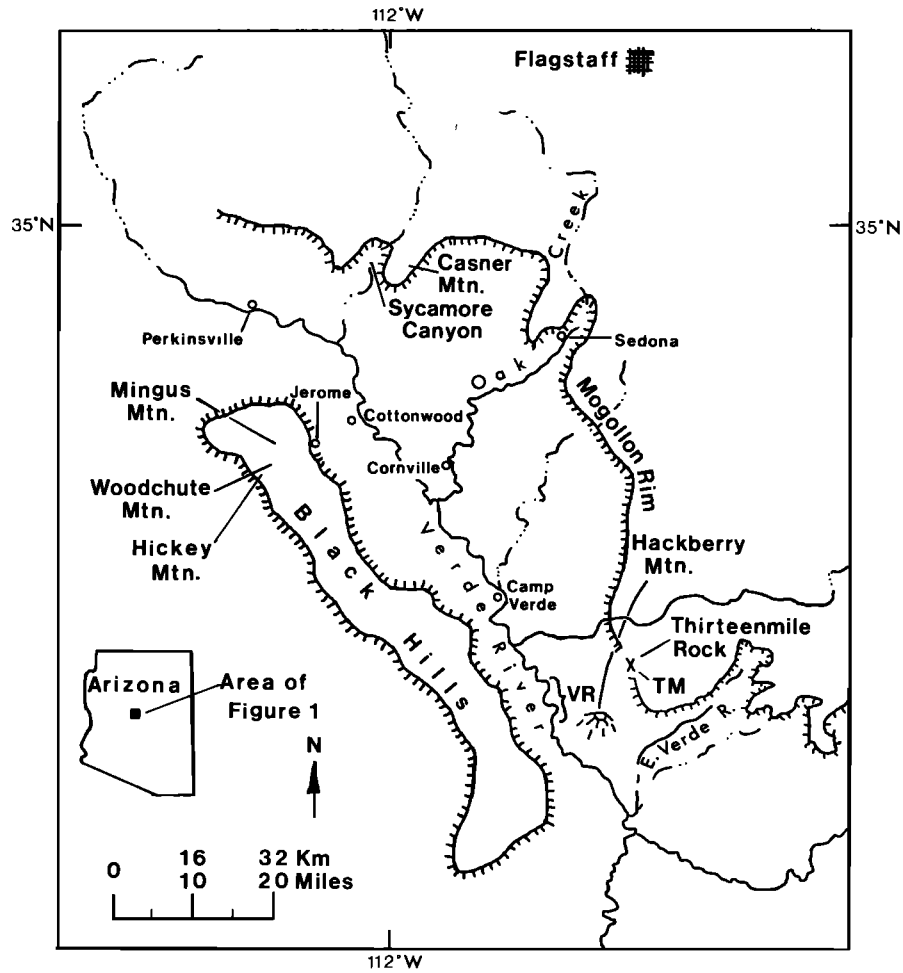


Fig. 1. The Hackberry Mountain–Verde Valley region of central Arizona, stratigraphic sections of Verde River (VR), and Thirteenmile Rock (TM).

berry Mountain volcano interbedded with basalt flows erupted from fissures and cones in the Hackberry Mountain area. The oldest dacite eruption occurred about 11.3 m.y. ago (unit T14, Figure 2), and the youngest flows in the Thirteenmile Rock section, northeast of Hackberry Mountain, were erupted between 3 and 5 m.y. ago. Silicic eruptions associated with the Hackberry Mountain volcano thus were erupted intermittently during a span of about 7 m.y. Stratigraphic relations between dacite, basalt, and sedimentary rocks north of Hackberry Mountain are diagrammed in Figure 3. Along the Verde River the basalt flow that directly underlies the conglomeratic strata of the Verde Formation served as a mapping datum. The underlying and overlying basalt (B) and tuff (T) beds are stratigraphically referred to this datum by number (Figures 2, 3, and 4). Minus numbers refer to strata above the datum.

METHODS OF STUDY

The study started with detailed geologic mapping in the Hackberry Mountain area (Figure 1) at a scale of 1:5000 to establish details of the stratigraphy. An area of about 60 km^2 was mapped for this purpose. When mapping was complete and a composite stratigraphic column determined (Figure 2), samples were collected from all mapped units for the determination of paleomagnetic directions and polarity. At the same time, samples from near the top and bottom of the section were dated by whole-rock K-Ar methods. It was assumed that the reversal pattern determined in the Hackberry Mountain area would serve as time control when evaluated with respect to the published sea floor reversal pattern for the interval of time between the dated rocks. Additional K-Ar age determinations would be made if needed.

PALEOMAGNETIC STUDIES

Samples were collected by means of a hand-held diamond-core drill similar to that described by *Doell and Cox* [1965]. Cores, 2.5 cm in diameter and 5–10 cm long, were oriented before being detached from the outcrop; the magnetic compass and clinometer were read to the nearest half degree, with an accuracy of about ±1°. For individual flows, 8–18 samples were obtained, commonly in pairs, across individual cooling units to ensure that representative paleomagnetic directions were obtained. Thin beds of sandstone separating flows were sampled across the sections to obtain possible paleo-secular variation information. Before orientations by compass were attempted, the outcrops were checked with the compass for abrupt variations in direction that might denote lightning strikes. In such situations, an offset magnetic orientation procedure or a solar transfer method was employed for core orientation. Wherever possible, flows were sampled in cliff faces and in outcrops that, for the most part, seemed not to be particularly vulnerable to lightning strikes.

After trimming of the cores, the natural remanent magnetization (NRM) was measured in an air-driven spinner magne-

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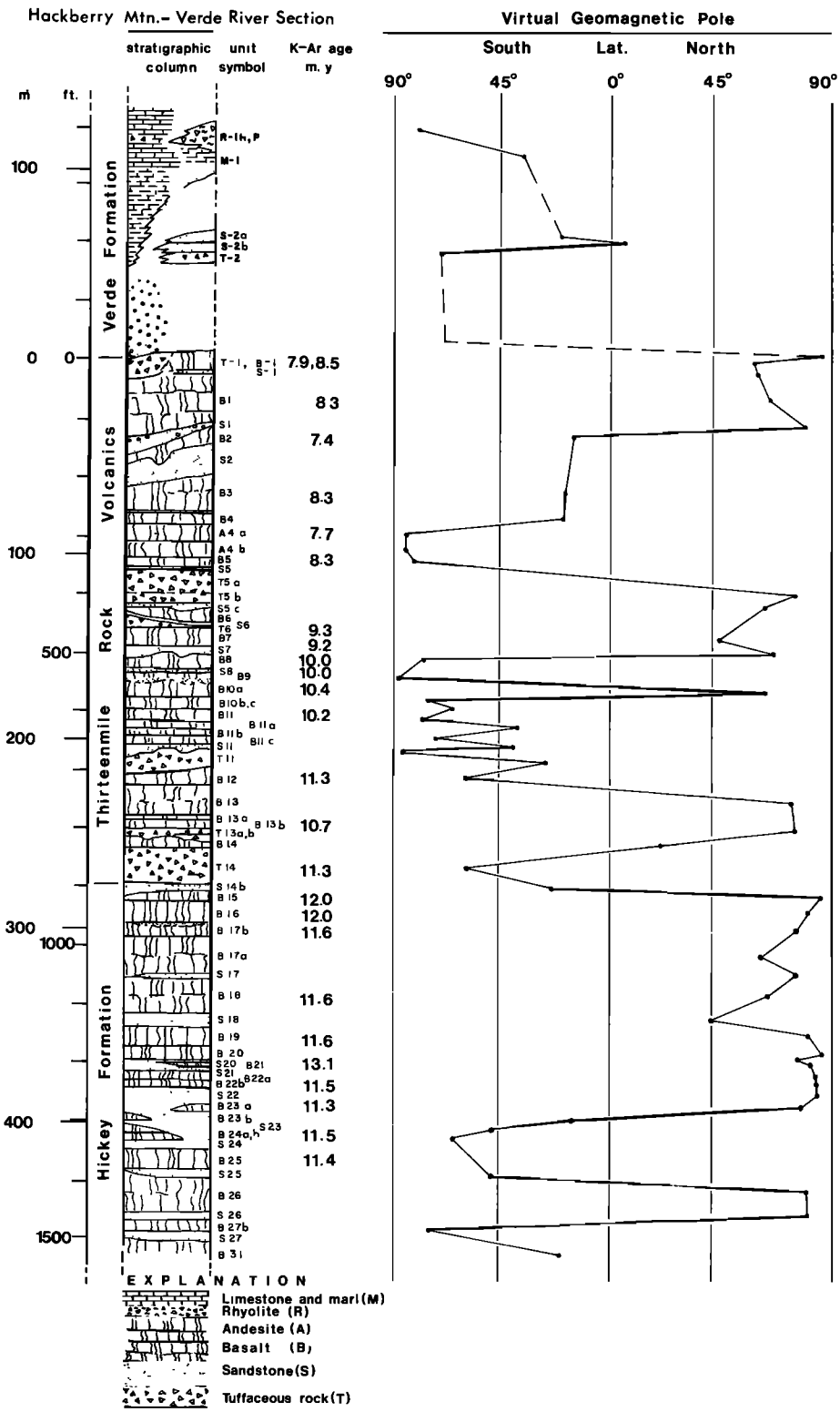


Fig. 2. Stratigraphic column northwest of Hackberry Mountain with unit designations, K-Ar ages, and latitude of virtual geomagnetic poles.

tometer having a noise level of about 2×10^{-7} emu. Pilot specimens for each group of cores then were subjected to progressive stepwise demagnetization analysis. Alternating field demagnetization was employed on samples from the volcanic strata, and both alternating field and thermal demagnetization analysis were employed on samples from the interbedded volcaniclastic sandstone. Criteria used to select samples exhibiting stable directions of magnetization included (1) minimum dispersions for groups of specimens from individual stratigraphic units, (2) magnetization vectors that trend toward the origins of orthogonal demagnetization diagrams [*Zijderveld*, 1967], and (3) the end points of demagnetization paths that

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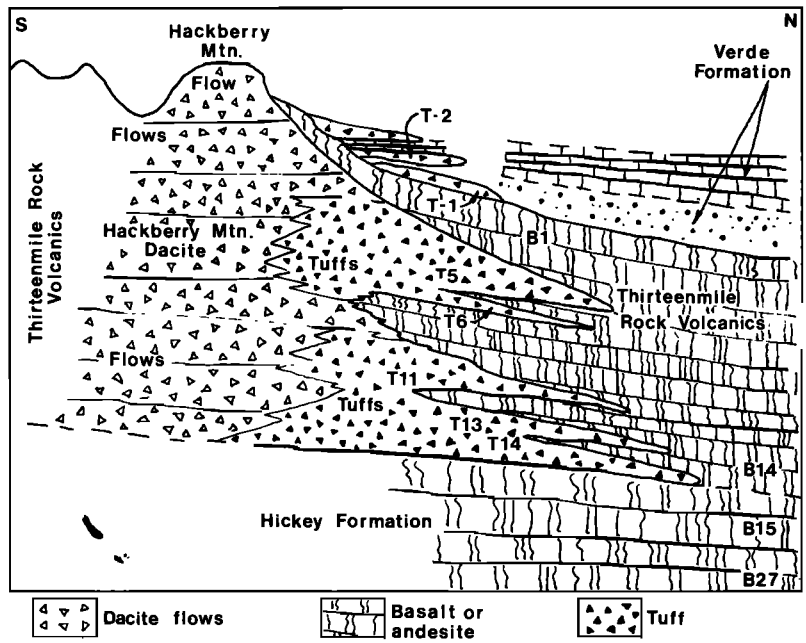


Fig. 3. Diagram of stratigraphic relations at southeast end of Verde Valley, North-central Arizona. For key to numbered units see Figure 2.

display shifts slightly to markedly away from the present field direction, reflecting removal of overprints attributable to weathering.

After the pilot specimens were partially demagnetized, the remaining specimens were cleaned at single or multiple demagnetization steps, depending on response to the cleaning. Individual specimens that failed to respond to demagnetization treatment and were anomalous in direction with respect to other specimens of the sample group were discarded. Specimens that allowed paleomagnetic directions to be determined were retained. Mean directions are summarized in Table 1.

The paleomagnetic directions in Table 1 are arranged downward in order of increasing stratigraphic age. Of 62 units sampled, 56 provided data for the calculation of paleomagnetic directions. Of these, two units of normal polarity and seven units of reversed polarity exhibit paleolatitudes less than 45° and were not used for the calculation of the mean normal and reversed polarity. Paleomagnetic directions for the individual stratigraphic units are uncorrected for structural tilt. In nearly all cases, a correction for tilt results in movement of the virtual geomagnetic pole away from the present field normal polarity direction or the (antiparallel) reversed polarity direction. This indicates that most, if not all, of the present-day attitude was acquired at the time of deposition and that the dips mainly are initial dips. Bedding dip may have been modified by tilting associated with faulting, but the amount is not determinable. Because many large and small gaps exist in the stratigraphic record, no attempt has been made to portray the paleomagnetic directions sequentially in a plot depicting paleosecular variation. No evidence of secular variation was observed within sandstone units. Paleomagnetic directions within the sedimentary units tended to cluster very well, reflecting either rapid deposition or the superposition of a single paleomagnetic direction after deposition.

The polarities of the individual units listed in Table 1 are plotted in Figure 2 as the latitude of the virtual geomagnetic pole. Most of the paleo poles lie far above or below the equator. A few lie about 20° or less from the equator. Reversed polarity directions are more scattered than directions from units of normal polarity, presumably because of incomplete removal of components of magnetization acquired during weathering in the present normal polarity field (compare the mean normal and reverse polarity directions for the Hackberry Mountain section in Table 1).

K-AR DETERMINATIONS

A total of 26 K-Ar age determinations were made for this study (Table 2): 22 whole rock and 4 mineral separates (3 biotite, 1 alkali-feldspar). The first age determinations were on samples of (1) tuff underlying the basal sandstone of the Verde Formation, and (2) a flow near the bottom of the volcanic sequence exposed along the Verde River. Later sampling and dating of nearly all of the intervening lava flows and some of the beds of tuff was found to be necessary in order to evaluate ages of individual flows and the lengths of time missing between flows. The 26 ages (Tables 2 and 3) are shown on the composite stratigraphic column (Figure 2) adjacent to the unit.

Sample preparation and argon and potassium analyses were carried out in the U.S. Geological Survey laboratories, Menlo Park, California. All samples except the first two (top and bottom flows) were collected, prepared, and analyzed during a period of 2 months under similar conditions to ensure uniformity. Argon-38 traces were a sequence from the same bulb. Potassium analyses were performed by a lithium metaborate flux fusion-flame photometry technique, the lithium serving as an internal standard [*Ingamells,* 1970]. Argon analyses were performed by standard isotope dilution procedures, using a 60° sector, 15.2-cm radius Neir-type mass spectrometer operated in the static mode for mass anlaysis. The amount of radiogenic ^{40}Ar from the different samples ranged from 2.4 to 51.5% with the variation at one standard deviation being large for samples containing less than 10% radiogenic ^{40}Ar. The reproducibility (±) ranged from 1.7 to 29% of the age (Table 2). Because most samples contain relatively small amounts of radiogenically produced ^{40}Ar with respect to contamination by

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atmospheric ^{40}Ar, the precision is too poor for close age designations of most of the individual samples. Because of the stratigraphic control, however, the actual age of a rock unit can be estimated more closely by considering the series of calculated ages (including their ±) within the framework of stratigraphic position. The values derived by evaluating analytical precision with respect to stratigraphic position are termed accuracy value on Table 3 and are shown as the shaded areas on Figure 4.

In order to evaluate the actual ages of the rock units, the K-Ar determinations were arranged in stratigraphic succession and shown as bars, the lengths of which represent the limits of analytical uncertainty at one standard deviation (Figure 4). Because the relative ages of all the rock units are known from the stratigraphic sequence, it is possible to evaluate the individual radiometric ages by comparing them with ages from overlying and underlying rocks. The lengths of many of the uncertainty bars become shortened because of constraints imposed by samples with comparatively small analytical errors (±). The shaded area in Figure 4 shows the age range for each of the samples. For example, sample B23a (third sample from the right, Figure 4) has a calculated K-Ar age of 11.3 ± 1.8 m.y. It underlies and is older than samples B22, B21, and B19, the last having a K-Ar age of 11.6 ± 0.2 m.y. Sample B23a must be older than 11.4 (11.6–0.2)m.y. We also know that it overlies and is younger than samples B24 and B25, the latter having a K-Ar age of 11.4 ± 0.3 m.y. Sample B23a can be no older than 11.7 (11.4 + 0.3) m.y. Its age therefore must lie in the interval between 11.4 and 11.7 m.y. (11.55 ± 0.15 m.y.), a considerable improvement over the K-Ar determination of 11.3 ± 1.8 m.y. The ages of the samples (accuracy value) listed on Table 3 have been ascertained by this method. The refined age estimates of the various units have an average age span of 4.4% of the midpoint value age with a maximum variation of 8.7% and a minimum variation of 1.3%. The age of the rock units cannot be estimated any more closely using the radiometric age and geologic constraints.

In addition to narrowing age evaluations for individual flow units in the sequence, the diagram (Figure 4) gives an indication of where major breaks in the depositional or time record exist. Five major hiatuses are recognized, represented by inverted steps (steps on the bottom edge) on the progression of evaluated ages. They occur between units A4a and B5, B5 and T6, B7 and B8, B11 and B12, and T13 and T14. In addition to recognition of these hiatuses the lengths of time represented at each hiatus can be estimated.

Magnetic Reversal Patterns

The first reversal chronology we produced in 1974 for the units in the Hackberry Mountain area. The pattern was adjusted to correspond to the measured stratigraphic column—reversed and normally magnetized intervals were aligned with the appropriate rock unit. Age control was two K-Ar age determinations, one near the top (tuff T-1, 7.7 ± 0.6 m.y.; the age determination in 1974 was based on different decay constants and abundance ratios than used in this study, see Table 3, T-1) and one near the bottom (lava flow B25, 11.1 ± 0.4 m.y.; the 1974 age based on old constants) of the section. The total time span of this sequence of rocks is thus about 3.4 m.y., and the number of measured polarity switches in the interval is 10. The comparable section between about 7.5 and 11.5 m.y. of the *Heirtzler et al.* [1968] marine reversal time scale was placed beside the Hackberry Mountain pattern, and an attempt was made to correlate or match normal and reversed intervals. A reasonable match emerged, in which a thick normal polarity interval in the Hackberry Mountain section was correlated with marine magnetic anomaly 5, having an estimated age of about 9–10 m.y. The Hackberry Mountain section has one additional interval of reversed polarity making a one-to-one match impossible, but the overall comparison seemed encouraging especially if short reversal switches had been missed in the pioneering study of Heirtzler and others.

The next step was to obtain tighter age control on the Hackberry Mountain rocks to verify or change the match in the interval between about 7.5 and 11.5 m.y. Twenty-four additional samples of lava flows and tuff between the original upper and lower units were collected for dating. The 24 intermediate rock units represented almost all the datable units in the section.

When the new ages were placed next to the appropriate rock units on the stratigraphic diagram (see Figure 2), it was obvious that there were periods of voluminous volcanism and rapid sedimentation, and periods represented by no rocks. The record thus is not one of uniform deposition. Rather, rates of emplacement were variable, and hiatuses of undetermined duration occur at many palces in the section. There is no simple relation between thickness of rock and time. The magnetic reversal diagram for strata in the Hackberry Mountain area thus needed to be adjusted to recognize time gaps and their lengths if it was to be related to a fixed time scale.

This was accomplished using the 27 age determinations, evaluated with respect to one another in stratigraphic context (Figure 5). More accurate ages were assigned to the units, unconformities were incorporated in the analysis, and the amounts of unrecorded time at the hiatuses estimated. In addition to the five main gaps revealed in the K-Ar evaluation diagram (Figure 4), others recognized by geologic mapping of features such as erosion, weathered surfaces, and sedimentary intervals are incorporated in Figure 5. From mapping, at least short hiatuses exist between nearly every volcanic unit in the sequence. Considering the hiatuses that can be recognized and in part evaluated with respect to time, it seems that for about half the time between 11.5 and 7.9 m.y. ago there is no volcanic or sedimentary record in the Hackberry Mountain area.

The evaluated magnetic reversal pattern from Hackberry Mountain is compared in Figure 6 with the most recently published ocean floor pattern of *LaBrecque et al.* [1977] and with the pattern from marine sediments reported by *Opdyke* [1972] and *Opdkye et al.* [1974].

The pattern from the Hackberry Mountain terrestrial sequence can be matched in general with the ocean floor pattern. The major magnetic anomaly in the time interval of 7.9–11.5 m.y. (anomaly 5 or polarity epoch 9) is correlated with a comparatively narrow stratigraphic interval of the Hackberry Mountain sequence. Polarity epochs 8, 10, and 11 also can be recognized, but the many short normal and reversed polarity switches within these epochs cannot be matched. These parts of the patterns can be made to correspond by shifting various segments of the Hackberry Mountain and ocean floor sequence by variable amounts. The older part (10–11.5 m.y.) of the Hackberry Mountain sequence would correspond well with the ocean floor patterns if the ocean floor pattern was considered to be about 0.2 m.y. older. Other discrepancies between the patterns could also be eliminated by changing the duration of the hiatuses in the Hackberry Mountain chronology within the framework of the existing temporal control.

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TABLE 1. Cleaned Paleomagnetic Directions for Flows and Interbedded Sandstone, Hackberry Mountain Volcanic Center, Central Arizona

Unit	Dip, deg	Latitude, deg	Longitude, deg	Treatment, Oe; °C	Number of Samples	Polarity*	Declination, deg	Inclination, deg	k†	$\alpha 95$‡, deg	Virtual Geomagnetic Pole: Latitude, deg	Virtual Geomagnetic Pole: Longitude, deg	δm§, deg	δp, deg
R-ln,p	3	34.433	−111.758	320	14	R	192.4	−54.8	55.9	5.4	−79.8	149.8	7.6	5.0
M-1	3	34.433	−111.767	15	6	(R)	143.8	15.5	33.5	11.7	−35.6	−65.6	12.1	6.2
S-2a	1	34.433	−111.783	30	5	(R)	236.1	15.3	301.0	4.4	−22.3	−174.5	4.5	2.3
S-2b	1	34.433	−111.783	30	3	(R)	254.1	51.1	15.8	32.2	6.0	−167.1	43.5	29.4
T-2	1	34.433	−111.783	160	6	R	199.7	−40.0	4.3	36.7	−69.3	−172.8	44.2	26.6
T-1	1	34.433	−111.783	130	18	N	359.1	53.5	33.8	6.0	89.1	130.9	8.4	5.9
B-1	4	34.433	−111.790	60	8	N	24.9	25.3	39.9	8.9	59.1	15.3	9.6	5.1
S-1	4	34.450	−111.790	50	4	N	29.6	36.3	549.4	3.9	60.2	−0.9	4.6	2.7
B1	4	34.450	−111.790	160	6	N	341.2	30.8	7.5	26.2	65.5	116.4	29.2	16.3
S1	4	34.450	−111.790	30	5	N	8.8	47.4	103.4	7.6	80.5	14.2	9.8	6.4
B2	3	34.450	−111.791	320	6	R	276.4	−56.0	34.4	11.6	−15.3	124.0	16.6	12.0
B3	3	34.450	−111.790	320	5	R	271.2	−57.1	138.1	6.5	−19.4	125.2	9.5	6.9
B4	3	34.450	−111.790	160	7	R	280.3	−67.2	77.9	6.9	−19.8	110.5	11.4	9.5
A4a	3	34.442	−111.767	320	8	R	183.0	−51.0	108.1	5.4	−86.3	−154.3	7.2	4.9
A4b	3	34.442	−111.767	160	9	R	180.6	−49.9	469.2	2.4	−86.2	−119.5	3.2	2.1
B5	6.5	34.450	−111.792	60	15	R	183.9	−44.4	85.1	4.2	−81.0	−134.8	5.2	3.3
T5b	6.5	34.450	−111.792	50	15	N	7.6	64.7	113.6	3.6	76.6	−88.8	5.8	4.7
S5c	7	34.450	−111.792	60	6	N	30.6	61.3	29.3	12.6	64.9	−49.5	19.4	14.9
B7	6.5	34.450	−111.792	160	7	N	40.0	81.0	16.4	15.4	46.9	−95.3	29.7	28.6
S7	6.5	34.450	−111.792	30	6	N	8.8	72.4	45.0	10.1	66.1	−100.2	17.9	15.9
B8	10	34.450	−111.792	60	7	R	166.8	−53.4	102.8	6.0	−79.1	−20.7	8.3	5.8
B9	6	34.450	−111.792	60	7	R	179.6	−53.6	42.6	9.4	−89.6	−64.4	13.1	9.1
B10a	11	34.440	−111.758	30	8	N	25.7	69.0	235.2	3.6	64.3	−74.3	6.2	5.2
B10b	11	34.433	−111.758	60	4	R	177.3	−44.0	110.0	8.8	−81.0	−95.9	11.0	6.9
B10c	9	34.442	−111.770	60	7	R	198.5	−64.5	41.1	9.5	−71.6	112.1	15.3	12.3
B11	11	34.440	−111.758	60	7	R	179.1	−45.5	293.0	3.5	−82.5	−105.7	4.5	2.9
B11a	12.5	34.430	−111.783	320	5	R	244.8	−59.8	2.3	65.4	−39.4	130.9	98.6	74.3
B11b	12.5	34.430	−111.783	160	6	R	174.4	−66.6	25.7	13.5	−74.7	54.3	22.2	18.3
B11c	12.5	34.425	−111.783	160	6	(R)	165.9	23.2	1.7	77.7	−41.6	−93.2	82.6	43.9
S11	11	34.440	−111.758	160	10	R	177.1	−54.7	148.3	4.0	−87.5	−2.9	5.6	4.0
T11	12.5	34.425	−111.783	60	7	R	87.7	−75.7	15.4	15.9	−29.3	36.9	29.3	26.9
B12	12.5	34.440	−111.758	160	7	R	147.5	−39.8	239.1	3.9	−59.3	35.8	4.7	2.8
T13a	12.5	34.433	−111.767	180	9	N	10.3	65.2	89.4	5.5	75.0	−83.6	7.2	4.8
T13b	12	34.430	−111.783	160	8	N	353.8	65.4	34.7	9.5	76.1	−129.5	15.5	12.5
B13	12.5	34.433	−111.765	50	8	N	13.3	65.0	484.2	2.5	74.0	−77.0	4.1	3.3
B14	12	34.425	−111.783	160	8	N	241.8	76.2	56.2	6.9	19.6	−136.1	12.8	11.9
T14	10	34.433	−111.767	50	9	R	190.7	76.2	57.7	8.0	−60.0	77.7	14.8	13.7
S14b	15	34.425	−111.775	130	11	R	75.4	−74.2	9.0	16.1	−22.9	37.1	29.2	26.4
B15	15	34.433	−111.767	64	11	N	356.7	53.7	19.2	10.7	87.3	155.9	12.9	7.8
B16	15	34.433	−111.767	32	8	N	9.2	53.1	85.9	6.0	82.3	−18.7	8.3	5.8
B17b	13	34.433	−111.767	60	9	N	354.6	66.0	500.3	2.3	75.6	−126.4	3.8	3.1
B17a	12	34.422	−111.780	610	13	N	326.5	55.4	91.5	4.4	62.7	171.0	6.2	4.4
S17	12	34.422	−111.780	60; 590	4	N	343.6	50.1	285.3	5.5	75.7	148.4	7.3	4.9
B18	15	34.422	−111.780	100	18	N	329.5	51.3	160.5	2.7	64.5	161.3	3.7	2.5
S18	24	34.422	−111.780	300	26	N	299.1	62.6	434.8	1.4	42.9	−171.0	2.1	1.7
B19	15	34.422	−111.780	60	10	N	348.8	54.3	325.4	2.7	80.8	163.9	3.8	2.7
B20	18	34.422	−111.780	60	8	N	359.4	56.2	77.3	6.3	87.6	−124.2	9.1	6.6
S20	18	34.422	−111.780	160	3	N	347.2	41.9	152.4	10.0	74.9	119.0	12.3	7.5
B21	18	34.422	−111.780	160	8	N	356.1	46.1	111.7	5.3	82.3	94.8	6.7	4.3

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B22a	15	34.422	−111.780	130	10	N	5.3	59.9	44.3	7.3	82.4	−80.0	11.1	8.4
B22b	15	34.422	−111.780	160	17	N	5.6	54.8	49.1	5.1	85.3	−35.0	7.3	5.2
S22	25	34.408	−111.808	60	9	N	358.9	48.8	71.7	6.1	85.2	79.6	8.1	5.3
B23a	15	34.422	−111.780	60	4	N	341.1	51.3	81.0	10.3	76.5	152.4	13.9	9.5
B23b	25	34.408	−111.808	320	8	(R)	116.3	18.1	1.6	71.4	−15.6	45.2	74.2	38.5
S23	25	34.408	−111.808	320	4	R	137.3	−31.0	13.4	26.0	−48.1	−35.7	29.1	16.3
B24a	25	34.408	−111.808	60	8	R	149.3	−58.5	66.8	6.8	−65.1	−1.7	10.1	7.5
B24b	15	34.422	−111.780	60	9	R	112.2	−65.3	164.9	4.0	−38.8	14.6	6.5	5.3
S25	16	34.408	−111.808	320	3	R	127.2	−60.2	20.4	28.1	−48.4	3.7	42.5	32.3
B26	16	34.408	−111.808	160; 320	8	N	5.5	42.4	63.8	23.8	79.0	40.8	29.3	18.0
S26	16	34.408	−111.808	130; 160	10	N	350.0	47.2	33.9	8.4	79.5	125.8	10.9	7.1
B27b	16	34.408	−111.808	160	7	R	162.7	57.6	159.3	4.8	−75.6	−1.6	7.0	5.2
B31	16	34.408	−111.808	320	5	(R)	79.9	−66.2	5.0	38.1	−19.1	24.6	62.5	51.8
Mean Directions[¶]					Number of Units		$\bar{D}$	$\bar{I}$	k	$\alpha 95$, deg	Virtual Geomagnetic Pole Latitude, deg	Virtual Geomagnetic Pole Longitude, deg	δm, deg	δp, deg
Normal polarity					29		0.2	55.3	26.9	5.3	88.6	−106.6	7.5	5.3
Reversed polarity					18		171.9	−54.3	24.6	7.1	−83.4	−16.6	10.0	7.0

*N = normal, R = reversed; (R) indicates unit of probable reversed polarity.
†$k = n - 1)/(n - R)$.
‡Semiangle of the cone of 95% confidence level.
§Semimajor and semiminor axes of the oval of 95% confidence level.
¶Calculated using units having VGP latitudes of 45° or greater.

Most changes would have to be of the order of 0.1–0.3 m.y. to bring the reversals into alignment. Because the uncertainty of the evaluated ages is as high as 0.85 m.y. and most are about 0.5 m.y. (Table 2), there is no objective criterion or basis for making these adjustments. The chronology for the Hackberry Mountain sequence is not accurate enough to warrant arbitrary age designations of less than about 0.5 m.y. Short magnetic anomalies are impossible to pinpoint with respect to age in the magnetic sequence in the Hackberry Mountain area. In short, the reversal chronologies are in permissible agreement but cannot be matched in detail because of analytical constraints.

The Hackberry Mountain pattern compares less well with the marine sedimentary core chronology of *Opdyke et al.* [1974] (see Figure 6). There are parts of the marine core–Hackberry Mountain pattern, however, that match where the ocean floor–Hackberry Mountain patterns do not. The better agreement between the Hackberry Mountain and the ocean floor polarity chronologies implies that the marine sediment record is less complete and accurate.

Discussion

Lack of correlation between different polarity chronologies reflects three fundamental problems: (1) lack of accurate dating, (2) assumptions about uniformity of the rates of sea floor spreading and the accumulation of volcanic and sedimentary rocks, and (3) lack of fine-scale resolution of the magnetometer record of ocean floor anomalies.

Chronology

Most of the Cenozoic chronology used in magnetic polarity time scales is based on marine magnetic anomalies whose ages are extrapolated from anomalies, the radiometric dates for which are no older than about 4.5 m.y. Because of analytical limitations, radiometric dates on rocks older than about 6.0 m.y. have an analytical uncertainty that is greater than the length of many magnetic reversals. This limitation was noted by *Dalrymple et al.* [1967, p. 169], who stated '... the dating errors will be large relative to the lengths of polarity epochs and much larger than the average event length .... It is apparent from the figure that polarity events will probably not be distinguishable from dating errors in rocks older than about 6 m.y.' First-order dates thus are limited to the young end of the geologic time scale, and second-order dating arrived at by relative means is used for older events. As this study points out, rocks in the age range of about 8–12 m.y. yield radiometric ages, which, even employing stratigraphic control, are not accurate enough to assign age limits to many of the reversals in this time interval.

The assumption that the chronology of the ocean floor magnetic pattern is correct is also questionable. The ages placed on this scale by extrapolation from a short period at one end of the scale (<1–4.5 m.y.) have no radiometric or analytically derived age control points during the rest of the Cenozoic time. This reflects the fact that radiometric ages have analytical uncertainties that represent increasingly long periods of time in older and older rocks. If rates of sea floor spreading are not constant, the long extrapolation based on a short time interval at the young end of the time scale is invalid.

Rates of Sea Floor Spreading, Sedimentation, and Volcanism

The basis for the assigned ages of the Cenozoic ocean floor polarity zonation is the assumption that spreading rates have

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TABLE 2. K-Ar Ages and Analytical Data of Volcanic Rocks, Hackberry Mountain–Verde River, Central Arizona

Field Number and Location	U.S.G.S Isotope Laboratory Number	Unit Name	Mineral Dated	K_2O, %	^{40}Ar rad, 10^{-11} mole/g	^{40}Ar rad, %	Apparent Age, m.y. ± σ
T-1	4I336	tuff	biotite	6.66	7.582	17.2	7.9 ± 0.5
B-1	4I283	basalt flow	w.r.	1.345 1.368	1.662	5.9	8.5 ± 1.3
B1	4I284	basalt flow	w.r.	1.140 1.150	1.374	3.3	8.3 ± 2.2
B2	4I558	basalt flow	w.r.	0.881	0.938	3.8	7.4 ± 1.9
B3	4I332	basalt flow	w.r.	1.856 1.858	2.226	8.4	8.3 ± 0.8
A4a	4I313	andesite flow	w.r.	1.848 1.867	2.070	7.4	7.7 ± 0.8
B5	4I314	basalt flow	w.r.	0.522 0.522	2.633	17.0	8.3 ± 0.5
T6	4I508	tuff	alkali feldspar	1.308	1.760	24.0	9.3 ± 0.4
B7	4I285	basalt flow	w.r.	0.566 0.571	0.754	21.8	9.2 ± 0.4
B8	4I3041	basalt flow	w.r.	0.893 0.894	1.294	2.4	10.0 ± 0.7
B9	4I299	basalt flow	w.r.	0.734 0.737	1.059	13.4	10.0 ± 0.7
B10	4I303	basalt flow	w.r.	0.945 0.960	1.435	10.9	10.4 ± 0.8
B11	4I297	basalt flow	w.r.	1.207 1.209	1.786	24.6	10.2 ± 0.4
B12	4I319	basalt flow	w.r.	0.641 0.646	1.046	4.0	11.3 ± 2.6
T13	4I486	tuff	biotite	6.60	10.163	17.8	10.7 ± 0.6
T14	4I333	tuff	biotite	6.05	9.855	15.0	11.3 ± 0.7
B15	4I315	basalt flow	w.r.	0.572 0.584	1.004	9.8	12.0 ± 1.3
B16	4I308	basalt flow	w.r.	0.971 0.977	1.684	10.8	12.0 ± 1.2
B17b	4I316	basalt flow	w.r.	0.995 0.983	1.657	41.2	11.6 ± 0.3
B18	4I300	basalt flow	w.r.	0.713 0.715	1.198	17.7	11.6 ± 0.6
B19	4I317	basalt flow	w.r.	1.482 1.463	2.468	51.5	11.6 ± 0.2
B21	4I556	basalt flow	w.r.	0.695	1.315	5.0	13.1 ± 2.2
B22	4I553	basalt flow	w.r.	0.610	1.017	15.4	11.5 ± 0.8
B23	4I335	basalt flow	w.r.	0.575 0.584	0.944	7.0	11.3 ± 1.8
B24	4I310	basalt flow	w.r.	1.463 1.468	2.455	31.0	11.5 ± 0.4
B25	4I286	basalt flow	w.r.	1.759 1.762	2.900	40.0	11.4 ± 0.3

Constants used in all age determinations are $\lambda_\epsilon + \lambda_\epsilon' = 0.581 \times 10^{-10}$ yr^{-1}, $\lambda_\beta = 4.962 \times 10^{-10}$ yr^{-1}.
$^{40}K/K_{total} = 1.167 \times .0^{-4}$ mole/mole.
The abbreviation w.r. denotes whole rock.

been constant or uniform in detail over a long interval of geologic time. If the rates change, the straight-line extrapolation from 4.5 m.y. to early Cenozoic time will not give the correct age. A general slowing, increase, or fluctuation in the rate will render the time scale inaccurate in detail. Because various check points often cited are based on relative age analysis tied to extrapolated spreading rates, they are, in themselves, no real check [see also *Blakely*, 1974].

The same argument applies to time scales based on estimated rates of sedimentation or deposition of sequences of volcanic flows. The accumulation of deep-sea sediments seems to offer the best chance for deposition of sediments at a steady rate over a long period of time, because variations due to local erosion presumably should not be important. In such a case, thickness would be proportional to time, and the extrapolated time scale from marine magnetic anomalies should correlate with the polarity zonation from the marine sedimentary column. Accumulation of volcanic rocks is less likely to be at a constant rate, and only in certain circumstances can the eruption of flows be assumed to have occurred uniformly. This assumption was made in magnetic polarity studies in Iceland by *McDougall et al.* [1976, 1977].

The marine sedimentary chronology of *Opdyke* [1972] and *Opdyke et al.* [1974] differs from the ocean floor chronology of *LaBreque et al.* [1977] as seen in Figure 6. It may be that the sedimentation rate in the marine core was not constant and that there are unrecognized unconformities.

Constant deposition is even less likely for a terrestrial volcanic sequence. The study in the Hackberry Mountain area has clearly shown that the sequence of lava flows, interbedded

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TABLE 3. K-Ar Age Determinations With Precision Values at σ and Estimated Actual Age Based on Geologic Relationships and Calculated Radiometric Age (See Figure 5)

Unit Designation	Age and Precision at σ (Reproducibility of Analysis)	Estimated Age Accuracy*
T-1	7.9 ± 0.5	7.4–8.4 m.y.
O-1	8.5 ± 1.3	7.4–8.5 m.y.
B1	8.3 ± 2.2	7.4–8.5 m.h.
B2	7.4 ± 1.9	7.4–8.5 m.y.
B3	8.3 ± 0.8	7.5–8.5 m.y.
A4a	7.7 ± 0.8	7.5–8.5 m.y.
B5	8.3 ± 0.5	7.8–8.8 m.y.
T6	9.3 ± 0.4	8.9–9.6 m.y.
B7	9.2 ± 0.4	8.9–9.6 m.y.
B8	10.0 ± 2.9	8.9–10.6 m.y.
B9	10.0 ± 0.7	9.3–10.6 m.y.
B10	10.4 ± 0.8	9.6–10.6 m.y.
B11	10.2 ± 0.4	9.8–10.6 m.y.
B12	11.3 ± 2.6	9.8–11.3 m.y.
T13	10.7 ± 0.6	10.1–11.3 m.y.
T14	11.3 ± 0.7	10.6–11.7 m.y.
B15	12.0 ± 1.3	10.7–11.7 m.y.
B16	12.0 ± 1.2	10.8–11.7 m.y.
B17b	11.6 ± 0.3	11.3–11.7 m.y.
B18	11.6 ± 0.6	11.3–11.7 m.y.
B19	11.6 ± 0.2	11.4–11.7 m.y.
B21	13.1 ± 2.2	11.4–11.7 m.y.
B22	11.5 ± 0.8	11.4–11.7 m.y.
B23a	11.3 ± 1.8	11.4–11.7 m.y.
B24	11.5 ± 0.4	11.4–11.7 m.y.
B25	11.4 ± 0.4	11.4–11.7 m.y.

Estimated using age determintion with it ± value and constraints imposed by super position of the units; see Figure 5.

tuffs, and sedimentary rocks was deposited at very different rates during the 8–12 m.y. span of time (see Figures 4 and 5). The thickness of units or groups of units in no way corresponds to lengths of time of emplacement. In the Hackberry Mountain area there were two periods, each of about 0.1 m.y., in which about one half of the total section was emplaced (about 11.5 m.y. ago, units B17b to B25, and about 8 m.y. ago, units T-1 to A4a) (see Fig. 2).

Interpretation of the Rock Record and the Marine Magnetic Anomaly Record

Besides the fluctuations in continuous deposition, unconformities exist that represent variable periods of nondeposition, erosion, or both. Although they are recognizable in the field, it is impossible to estimate the amount of time represented by the unconformities. A change in magnetic polarity in two stratigraphically adjacent units tells nothing about the age difference between the units. A series of reversals also tells little about the total amount of time represented by, or missing from, the record. Inspection of the many magnetic chronology patterns from sea floor marine sediment or terrestrial rocks shows that reversals are not periodic or of equal length. Some intervals of single polarity last for a million years, others are less than a tenth as long.

The ocean floor magnetic reversal pattern is also subject to interpretation and not completely objective. Short events are difficult to distinguish, especially if they occur near the end of a long polarity event. Although the chances of finding a reversal in a terrestrial sequence that is too brief to have been recognized on the ocean floor record is not great, the possibility exists, and the result of this would make comparison of the patterns even more difficult.

SUMMARY

1. Terrestrial sequences of interbedded volcanic and sedimentary rocks are not suited for extending and dating the magnetic reversal time scale beyond the 4.5-m.y. age established by *Cox et al.* [1963, 1968] and *Cox* [1969]. This defi-



Fig. 4. Age evaluation diagram. Ages are placed in stratigraphic order, length of bar represents spread of the ± at σ of analytical uncertainty, shaded area delimits maximum possible limits of the ages ascertained by overlap of ± when arranged in stratigraphic superposition. Arrows mark unconformities.

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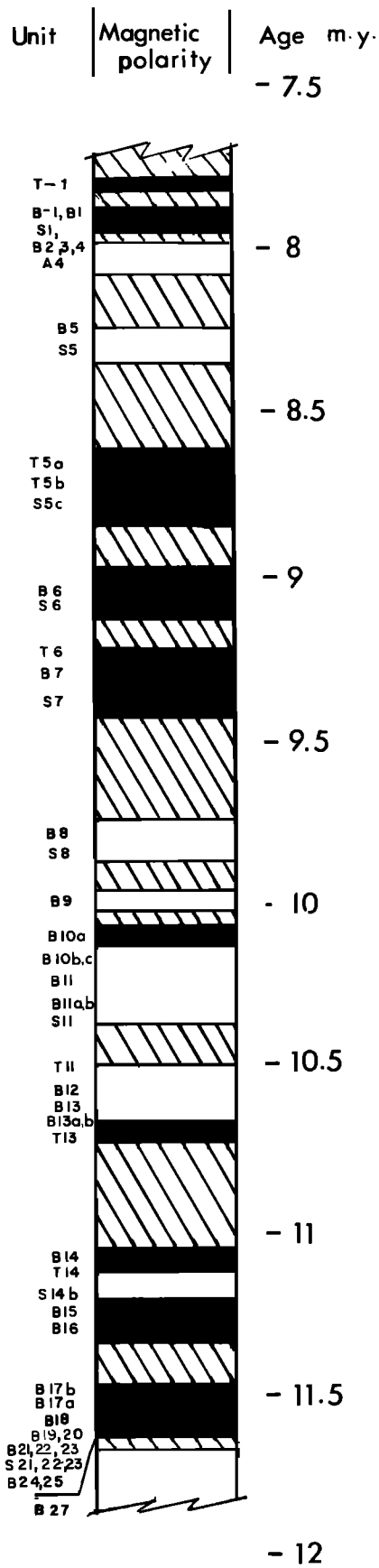


Fig. 5. Hackberry Mountain magnetic polarity pattern based on evaluated K-Ar dates and field evidence of unconformities. Unconformities are shown by crosshatched pattern.

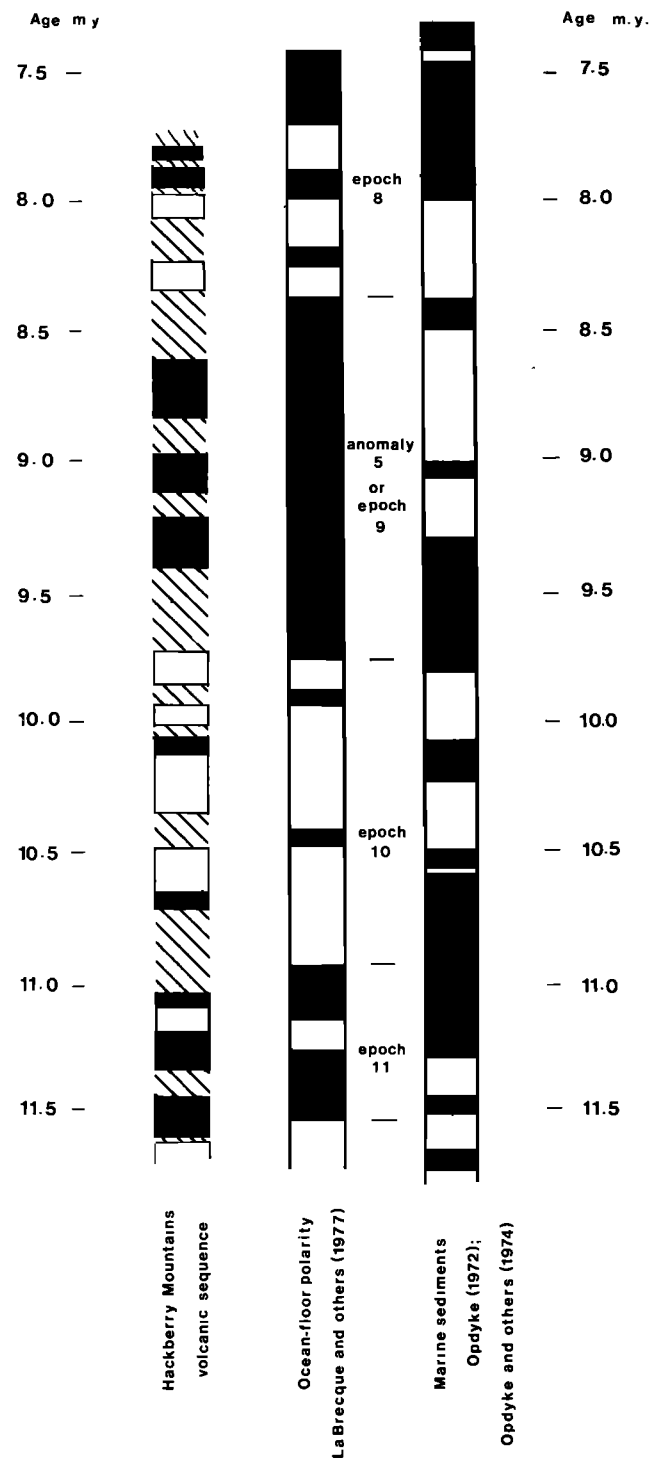


Fig. 6. Comparison of Hackberry Mountain evaluated and adjusted polarity zonation with polarity zonation reported from the ocean floor [*LaBrecque et al.*, 1977] and marine sediments [*Opdyke*, 1972; *Opdyke et al.*, 1974]. Cross-hatched areas are periods of no geologic (and magnetic) record based on evaluation of K-Ar ages and geologic relationships.

ciency is attributed to two causes: (1) difficulty in identifying and establishing the length of hiatuses or of intervals with unequal rates of deposition, and (2) analytical constraints that do not allow precise enough radiometric dating to distinguish short reversals.

2. The reversal pattern of a terrestrial sequence, regardless

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of the number of reversals present, is of little value in dating a sequence of rocks. Two rocks with different polarity can be said to be of different age, but their absolute age is not established. The reversal pattern cannot refine the date of the sequence beyond that established by radiometric dating or other means. Attempts to match a series of reversals with the ocean floor pattern and hence determine an age are of no value in the interval of 7–12 m.y. and probably for a much longer period of time than this. A polarity reversal pattern from terrestrial lavas by itself cannot be used to date the rocks.

3. There is no detailed correlation in the interval 7.5–11.5 m.y. among the various polarity time scales that have been proposed. The several polarity time scales derived from the ocean floor stripe pattern are in only moderate agreement (compare, for example, *Heirtzler et al* [1968], *LePichon et al.* [1973], *Blakely* [1974], *Tarling and Mitchell* [1976], *Watkins* [1976], *McDougall et al.* [1976, 1977], *LaBrecque et al.* [1977]). These, in turn, do not correspond well with the polarity record from marine sediments [*Opdyke,* 1972; *Opdyke et al.,* 1974]. The comparatively well-dated intervals containing polarity switches in the Hackberry Mountain section are in general agreement with the ocean floor record.

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